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Deep Sea Research Part II: Topical Studies in Oceanography

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Progression of marine heat wave events over the tropical Indian Ocean and its underlying mechanisms

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Highlights

- A Significant increase in number of MHW days and frequency is observed over the TIO for the period 2011–2021.
- Mixed layer heat budget reveals the dominant role of net heat flux and vertical processes in governing the MHW events.
- Subsurface warming caused by downwelling planetary waves reinforces the sustenance of MHW events.

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Abstract

Marine Heat Waves (MHW) are devastating extreme oceanic events that have severe and destructive effects on the marine and coastal ecosystems. In the present study, the generation and advancement of MHW events during the last four decades over the tropical Indian Ocean (IO) and its primary regulating factors are investigated. Multiple MHW events have been detected over the tropical IO in the past decade. It was found that between 2011 and 2021, the tropical Indian Ocean observed a significant increase in both the mean annual number of MHW days and the frequency of MHW occurrences. This coincided with a substantial rise in sea surface temperatures (SST) in the region during the same period. Long-running events were detected for the years 2015 over the Somali coast, 2016 over the Java-Sumatra Coast, and for 2019, and 2020 over the Seychelles Dome. Event-specific analysis revealed that a decline in wind speed was observed during the second phase of the 2015 MHW event along the Somali coast which resulted in the subsidence of upwelling, similar observations were also made around the Java-Sumatra coast. Subsequently, the role of planetary waves in the sustenance of the long-running MHW events is analysed. Positive sea level anomaly values were observed around Somali and Sumatra for 2015 and 2016, which in turn signify the presence of downwelling planetary waves. These planetary waves play an important role in oceanic surface and sub-surface warming and mixing by deepening thermocline and consequentially inhibiting the upwelling and entrainment. An analysis of the mixed layer heat budget terms over the active regions of MHW events in 2015, 2016, 2019, and 2020 shows that the primary contributor influencing these MHW events is net heat flux over the majority of the affected areas which is followed by the dominant role of vertical advection over the Somali region.

Keywords

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Introduction

One of the important effects of climate change is the rise in the number and severity of extreme weather events (IPCC, 2023). One such extreme event that can occur at any time of the year is a marine heat wave, which is defined as a contiguous expanse of exceptionally warm sea surface temperatures (SST) enduring for days and can last up to months (Frölicher et al., 2018). Outbursts of MHWs have reported being occurred in different regions of the world, e.g., over the Mediterranean Sea in the summer of 2003 due to the European heat waves (Olita et al., 2007; Ibrahim et al., 2021), the South China Sea and the Bering Sea (Yao and Wang, 2021; Carvalho et al., 2021), as well as major ocean basins like the Pacific, Atlantic, Indian, and Arctic Oceans (Joh and Di Lorenzo, 2017; Manta et al., 2018; Lenanton et al., 2017). The largest MHW event was recorded over the northeast Pacific Ocean between 2013 and 2015 with maximum SST anomalies of 6°C, due to the decline in low cloud fraction and surge in downward shortwave radiative fluxes (Peterson et al., 2015; Freeland and Whitney, 2014; Schmeisser et al., 2019). The prolonged weakening of high-pressure systems reduces wind speed, decreases evaporation cooling and wind-driven upper ocean mixing (Amaya et al., 2020; Vogt et al., 2022); an increase of net air-sea heat flux with positive horizontal advective heat flux anomaly (Maggiorano et al., 2021; Gao et al., 2020), intensification of ocean currents (Pearce and Feng, 2013), and anomalous latent heat flux (Schlegel et al., 2021) are responsible for the occurrences of MHW events. The bulk of MHWs in Indo-Pacific regions is associated with climatic modes such as the Pacific Decadal Oscillation, El Niño, and the Indian Ocean Dipole (IOD) (Saranya et al., 2022). MHW events are growing more frequent and intense with time. In the current scenario, 87 percent of MHWs are linked to human-induced warming, within the 1.5°C warming settings, MHWs are expected to become decadal to centennial events (Frölicher et al., 2018, Laufkötter et al., 2020). Within 3°C warming conditions, annual to decadal events in the preindustrial climate, MHWs only appeared once per hundreds to thousands of years (Frölicher et al., 2018, Laufkötter et al., 2020). Between 1925 and 2016, the frequency of MHW occurrences grew by 34%, their duration by 15%, and the estimated number of MHW days worldwide increased by 54%. (Oliver et al., 2018). Due to the enormous proliferation of extreme marine heat wave occurrences during the past century, several regional climates in various economically and ecologically crucial maritime regions have transitioned outside of their historical temperature ranges (Tanaka and Van Houtan, 2022). Several maritime regions will achieve a nearly constant MHW condition by the turn of the 21st century worldwide (Oliver, 2019). Therefore, a greater understanding of the effects of MHW and the factors governing their existence is necessary.

Saranya et al. (2022) explored the origin and evolution of MHWs in the Indian Ocean (IO) from 1982 to 2018 and their impact on the Indian summer monsoon. They stated that the western IO had the highest increase in MHWs, at 1.2–1.5 events per decade, followed by the northern Bay of Bengal at 0.4–0.5 events per decade. Past research has indicated the influence of rapid warming of the IO, El Niño events, and the presence of downwelling oceanic planetary waves on MHW occurrences (Zhang et al., 2021; Saranya et al., 2022; Dinesh et al., 2023). Azarian et al. (2024) studied MHWs in the southern IO and discovered that MHWs were linked to a reduced likelihood of encountering winter waters, which are characterized by a subsurface layer of minimum temperature. An analysis of ocean warming and MHWs, based on historical observations and CMIP6 climate projections, revealed intensified surface warming and MHW occurrences between 40°S and 55°S in the Antarctic Circumpolar Current region through the 21st century (Azarian et al., 2024). Dinesh et al. (2023) also studied MHWs and marine cold spells (MCSs) in the tropical TIO and observed that in the past two decades (2001–2010, 2011–2020), there has been a notable increase in the frequency and duration of MHWs during the JJAS months. The Indian Ocean Dipole (IOD) and the number of MHW days observed in the central Bay of Bengal during the IOD's autumn peak have a positive correlation, according to recent research, having a 95%

confidence level (Liang et al., 2024). Kadam et al. (2024) discussed the impact of IO's MHW on microalgae. Zinke et al. (2015) discussed the interaction between the MHWs of southeast IO and the western Pacific warm pool and their effect on corals.

The Indian Ocean (IO) is one of the most dynamic oceans because of the unique reversal of wind patterns (Dietrich, 1973), monsoon winds having a strong influence on the productivity of the western IO (Kumar et al., 2001), and the massive freshwater inflow in the northern IO (Saraswat et al., 2023). In spite, of the unique dynamics, there are limited studies of MHW over the IO. Since MHWs in this area may have an impact on fisheries, coastal populations, and marine biodiversity, we must comprehend them. Research and monitoring efforts are therefore needed to better understand and predict these phenomena and their impacts on the IO ecosystem and coastal communities. In this study, the occurrences of MHW events over the tropical IO (30°E – 120°E, 30°S – 30°N) are identified and investigations are carried out on the duration, and frequency of MHW events during the period from 1982 to 2021. The role of planetary waves on the occurrences of MHW events are also studied. Mixed layer heat budget analysis is carried out over multiple regions of interest to discern the relative contributions of the ocean dynamics and thermodynamics in the evolution of the MHW events.

Section snippets

Data

The MHW characteristics over the IO were studied using the Optimum Interpolation Sea Surface Temperature (OISST) which has a spatial resolution of 0.25°. The OISST data is built based on observations from multiple sources ranging from satellite-derived products to in-situ observations from various buoys and research vessels (Reynolds et al., 2007). For the present analysis, the OISST data is used over the period from January 1982 through December 2021. Chlorophyll-a data obtained from the ...

Advancement of MHWs over tropical IO

To understand the multi-decadal variability and decadal trend of SST over the IO, the area averaged SST anomaly for the tropical IO (30°S – 30°N, 30–120°E) is estimated for the period 1982–2002, as shown in Fig. 1. It can be seen from the figure that the tropical IO has experienced a phase change from a predominantly cooling phase to a warming phase in the recent decades. The strongest negative SST anomaly (−0.78°C) was recorded in 1987. Through the period of 1982–1998, the magnitude of ...

Conclusions

MHWs are one of the catastrophic climatic phenomena that can have significant adverse effects on the marine ecosystem and the regional as well as ecological and socio-economic stability. In this study, the generation and development of MHW events in the past four decades over the tropical IO is detected using OISST. It was found that several MHW events had occurred between 2010 and 2021 over the tropical IO with varying intensity and duration. These events mainly took place over the Arabian ...

CRedit authorship contribution statement

Diya Das: Writing – original draft, Visualization, Methodology, Formal analysis. **Sachiko Mohanty:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. ...

Declaration of competing interest

None. ...

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